6/11

21+3

EE2-08C Vector Calculus CLASS TEST 24 C

Please answer each question by clearly circling your answer, i.e. circling (a), (b), (c), (d) or (e). Note that some of the questions do not have a choice (e). Each correct answer will earn 4 marks. Each incorrect answer will receive -1 marks. A question left blank will receive 0 marks.

PLEASE WRITE YOUR NAME AND STUDENT NUMBER IN THE BOX BELOW:

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1. The volume $\mathcal{V}$ is the cube with one corner at the origin and another at the point $(-1, 1, 1)$. The (closed) surface of $\mathcal{V}$ is denoted by S. Define $\boldsymbol{G} = x^3y^2z^2\boldsymbol{i} - 2x^2y^3z^2\boldsymbol{j} + x^2y^2z^3\boldsymbol{k}$ and

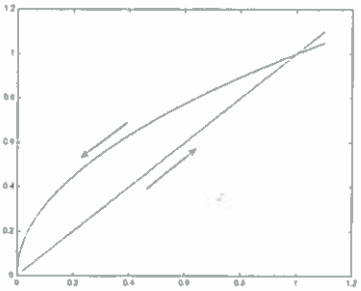
$$I = \iint_S \boldsymbol{G} \cdot \mathrm{d}\boldsymbol{S}\,.$$

Then the value of I is given by:

(a) $2\boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$, (b) 2, (c) 4π , (d) 0

(e) None of the above

2. The figure below shows a closed contour C, given by the line $y = x$ from the origin to the point $(1, 1)$, and the curve $x = y^2$ from $(1, 1)$ back to the origin.



The value of the line integral over this contour: $\oint_C (5x^2 + 2y^3)\,dx + (6xy^2 - x^3)\,dy$ is

(a) $\frac{3}{20}$, (b) $-\frac{3}{14}$, (c) $-\frac{3}{28}$, (d) $-\frac{3}{10}$,

(e) none of the above.

3. Consider the function $\phi(x, y, z) = \frac{x^2y}{z^3} + xe^{-z}$ at the point $P(1, 0, -1)$. What is the value of $\nabla\phi$ at P?

(a) $e + 2$, (c) $\frac{1}{e}\boldsymbol{i} + \boldsymbol{j} - e\boldsymbol{k}$, (b) $\frac{2}{e} - 1$, (d) $e\boldsymbol{i} - \boldsymbol{j} + e\boldsymbol{k}$,

(e) None of the above

4. Define $\boldsymbol{G} = (2y^2 - \sin x)\boldsymbol{i} + 4xy\boldsymbol{j} + (1-2z)\boldsymbol{k}$, and $\boldsymbol{G} = \nabla\phi$, with $\phi(0,0,0) = 0$. Then the functional form of ϕ is given by:

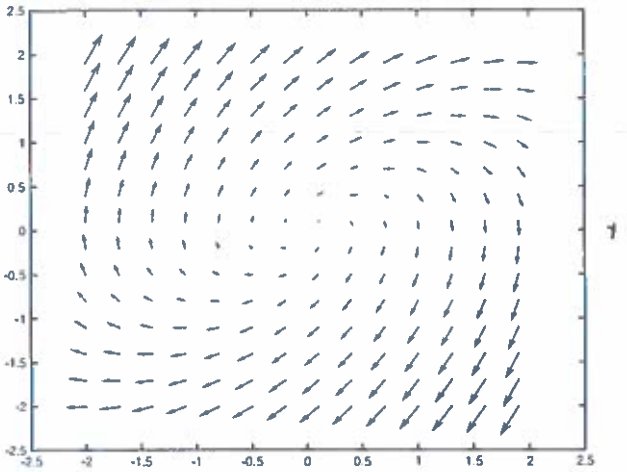
(a) $2xy^2 + \cos x - z(1-z)$, (b) $(2xy^2 + \cos x)\boldsymbol{i} + 2xy^2\boldsymbol{j} + z(1-z)\boldsymbol{k}$,

(c) $2xy^2 - \cos x + z(1-z)$, (d) ϕ does not exist, as $\boldsymbol{G}$ is curl-free,

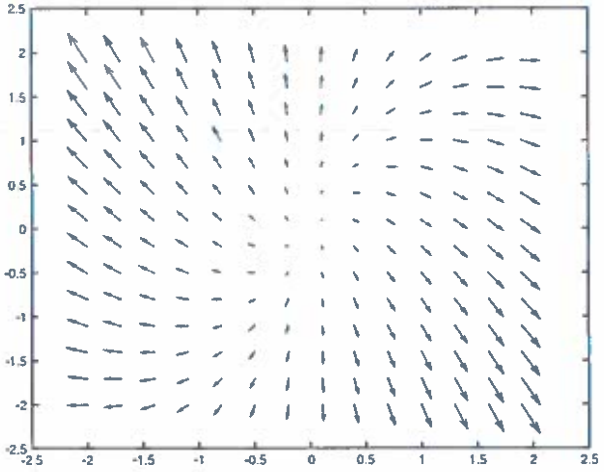
(e) None of the above

5. Which of the following figures shows the 2-D field

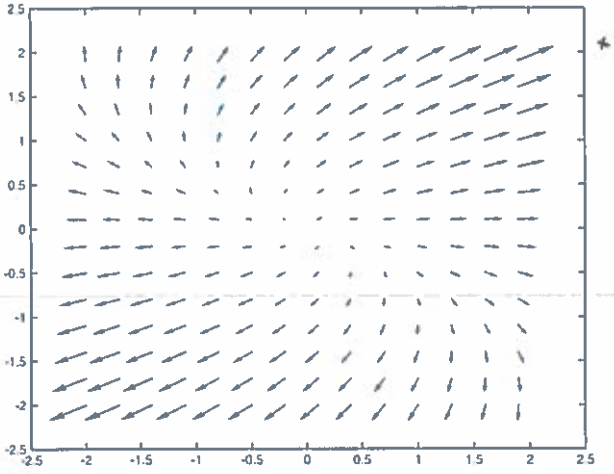
$$\boldsymbol{F} = x\boldsymbol{i} + (x+y)\boldsymbol{j}\,?$$



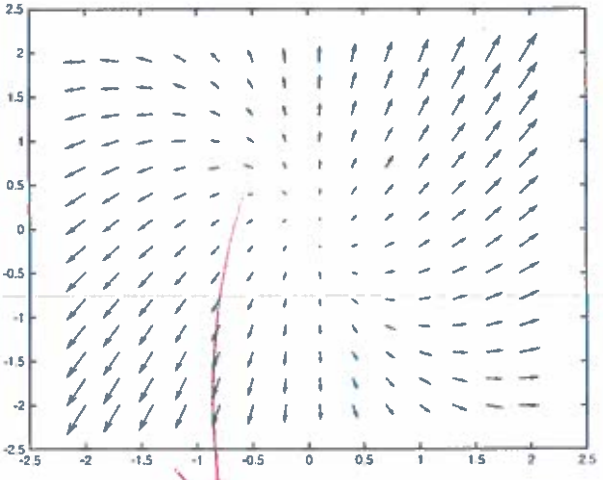
(a)



(b)



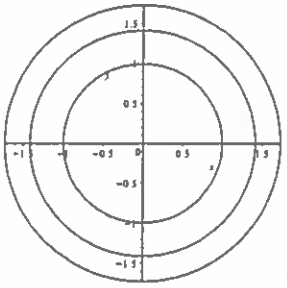
(c)



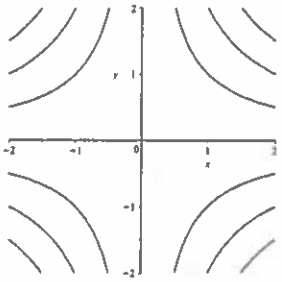
(d)

6. Which of the following figures shows the field lines for

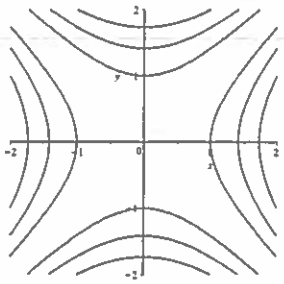
$$F = yi - xj\,?$$



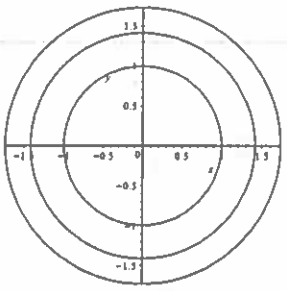
(a) (clockwise)



(b)

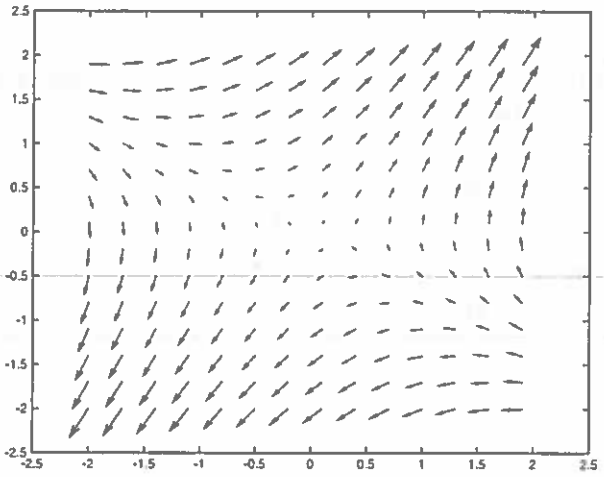


(c)



(d) (anti-clockwise)

7. Which of the following 2-D fields is shown illustrated in the figure below?



(a) $F = (y-x)i + xj$, (b) $F = xi + (x-y)j$, (c) $F = yi + (x+y)j$, (d) $F = (x+y)i + yj$,

8. Consider the line integral $I = \int_A^B ds$, along the anticlockwise curve $x^2 + y^2 = 1$, from $A = (0, -1)$ to $B = (0, 1)$. The value of I is given by:

(a) 0, (b) π, (c) $-\pi$ (d) 2π,

(e) None of the above

9. Consider the curve $\boldsymbol{r} = t\boldsymbol{i} + t^3\boldsymbol{j} + t^2\boldsymbol{k}, \quad (0 \le t \le 1)$. Define the function

$$\boldsymbol{F} = (y - 2xz)\boldsymbol{i} + 2x\boldsymbol{j} + (z - 2x^2)\boldsymbol{k},$$

and the integral $I = \int_A^B \boldsymbol{F} \cdot d\boldsymbol{r}$, where A corresponds to $t = 0$ and B corresponds to $t = 1$. The value of the integral I is:

(a) $\frac{1}{2}\boldsymbol{i} + \boldsymbol{j} - \frac{3}{4}\boldsymbol{k}$, (b) $\frac{1}{4}$, (c) $2\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$, (d) $\frac{3}{4}$,

(e) None of the above

10. Consider the vector field $\boldsymbol{F} = (y^2 - \sin x)\boldsymbol{i} + xy^2\boldsymbol{j} + (5 - z^3)\boldsymbol{k}$, and

$$I = \oint_C \boldsymbol{F} \cdot d\boldsymbol{r},$$

where C is the unit circle: $x^2 + y^2 = 1$. Then the value of I is given by

(a) $\frac{\pi}{2}$, (b) 0, (c) $-\pi$, (d) $-\frac{\pi}{2}$

(e) None of the above

Green's Theorem in a plane. Let $P(x, y)$ and $Q(x, y)$ be arbitrary, continuous functions in the $x - y$ plane, in which there is a closed, piecewise smooth boundary C enclosing a region R. Then

$$\oint_C (P dx + Q dy) = \int\int_R (Q_x - P_y)\, dx dy.$$

Divergence (Gauss') Theorem Given an arbitrary 3D vector field $\boldsymbol{u}(x, y, z)$ in a finite, simply connected volume V with closed surface is S, where dS is an element of area on S, then

$$\int\int\int_V \operatorname{div} \boldsymbol{u}\, dV = \oiint_S \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{S} = \oiint_S \boldsymbol{u} \cdot \hat{\mathbf{n}}\, dS$$

where $\hat{\mathbf{n}}$ is a unit vector normal to the surface S.

Stokes' Theorem Given an arbitrary 3D vector field $\boldsymbol{v}(x, y, z)$ and a surface S with boundary C,

$$\oint_C \boldsymbol{v} \cdot d\boldsymbol{r} = \int\int_S \operatorname{curl} \boldsymbol{v} \cdot \hat{\mathbf{n}}\, dS = \int\int_S (\boldsymbol{\nabla} \times \boldsymbol{v}) \cdot \mathrm{d}\boldsymbol{S}$$

where $\hat{\mathbf{n}}$ is a unit vector normal to the surface S.

Formulae

$$
\begin{aligned}
\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} &= a_x b_x + a_y b_y + a_z b_z \,, \\
\underline{\mathbf{a}} \times \underline{\mathbf{b}} &= (a_y b_z - a_z b_y)\underline{\mathbf{i}} + (a_z b_x - a_x b_z)\underline{\mathbf{j}} + (a_x b_y - a_y b_x)\underline{\mathbf{k}} \\
\nabla &= \frac{\partial}{\partial x}\underline{\mathbf{i}} + \frac{\partial}{\partial y}\underline{\mathbf{j}} + \frac{\partial}{\partial z}\underline{\mathbf{k}} \,. \\
\nabla\phi &= \frac{\partial \phi}{\partial x}\underline{\mathbf{i}} + \frac{\partial \phi}{\partial y}\underline{\mathbf{j}} + \frac{\partial \phi}{\partial z}\underline{\mathbf{k}} \\
\nabla \cdot \underline{\mathbf{F}} &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \,, \\
\nabla \times \underline{\mathbf{F}} &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\underline{\mathbf{i}} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\underline{\mathbf{j}} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\underline{\mathbf{k}} \\
\nabla^2\phi &= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \,. \\
\nabla \cdot (\phi \underline{\mathbf{F}}) &= \underline{\mathbf{F}} \cdot \nabla\phi + \phi \nabla \cdot \underline{\mathbf{F}} \\
\nabla \cdot (\phi \nabla\psi - \psi \nabla\phi) &= \phi \nabla^2\psi - \psi \nabla^2\phi \,, \\
\nabla \times \nabla \times \underline{\mathbf{F}} &= \nabla(\nabla \cdot \underline{\mathbf{F}}) - \nabla^2 \underline{\mathbf{F}}
\end{aligned}
$$